

Nitish Ravisankar Raveendran

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Education

University of Maryland, College Park

Master of Engineering in Robotics

Expected Graduation: Spring, 2025

Relevant Coursework: Advanced Robotics, Machine Learning, AI in Robotics, Sensor Integration, Computer Vision, Sensor Fusion

Anna University, PEC, Chennai

Bachelor of Engineering in Electronics and Communications Engineering

July 2022

CGPA: 8.94/10.0

Relevant Experience

Robotics Club, PEC Chennai *President*

March 2019 - August 2022

- Championed club initiatives that increased membership and orchestrated multiple innovative projects and events, engaging a large number of students.

Institute of Innovations, Chennai *Intern*

May - June 2021

- Designed and deployed IoT devices using ESP8266 and ESP-32, improving real-time data acquisition by 20%.

- Integrated cloud-based analytics, reducing latency in sensor data processing.

Projects

Bipedal Locomotion Using Reinforcement Learning

Fall 2024

- Implemented **TRPO**, **PPO**, and **TQC** for bipedal locomotion in simulation, optimizing efficiency and stability.

- Compared training performance across different **RL** algorithms to refine gait adaptation.

Gesture-Based Control in Assistive Technology

Fall 2024

- Developed a real-time hand-gesture recognition system using **Google MediaPipe** and **ROS2**.

- Enhanced TurtleBot teleoperation efficiency by 30% through **adaptive control algorithms**.

Multi-Agent Navigation with MCTS

Fall 2024

- Evaluated centralized vs decentralized navigation strategies in multi-agent robotic systems.

- Integrated **Monte Carlo Tree Search (MCTS)** in **ROS2** and **Gazebo**, improving path efficiency.

Lunar Light Utility Vehicle (LLUV)

Fall 2024

- Conceived and built a **lightweight, autonomous lunar rover** for site preparation and infrastructure support.

- Integrated adaptive suspension and refined wheel designs, achieving enhanced navigation efficiency on **simulated extraterrestrial terrains**.

Arduino-based Aimbot with YOLO Object Detection

Summer 2024

- Engineered an advanced Arduino-based aiming solution that incorporated **YOLO** technology to achieve over 95% accuracy in target recognition while streamlining the servo adjustment process, resulting in reduced setup times across projects.

Quadcopter Design and Development

Summer 2024

- Devised and constructed an **autonomous quadcopter** by integrating mechanical, electronic, and software components, enhancing control responsiveness.

- Implemented simulation and control systems in **Python** and **C++** to streamline flight performance.

Reinforcement Learning for Robot Pursuit and Evasion

Spring 2024

- Devised a reinforcement learning framework using **DQN** for training robot pairs (**evader and pursuer**) in a 3D **ROS2** and **Gazebo** environment, bolstering strategic decision-making.

Alpha - Pick and Place Mobile Manipulator

Fall 2023

- Innovated and crafted a **CAD-based** mobile manipulator integrated with **ROS2**, mitigating operational errors during precision pick-and-place tasks.

Technical Skills

Programming: Python, C++, MATLAB

Machine Learning: PyTorch, TensorFlow, Deep Reinforcement Learning

Robotics: ROS2, Gazebo, Computer Vision, Perception Systems

Embedded Systems: Arduino, ESP8266, ESP-32

Simulation and Algorithms: Sensor Fusion, State Estimation, Depth Estimation, Path Planning

Tools: JAX, Docker, Git